

Ben Gu

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA

M.S. in Robotics *GPA: 4.0/4.0*

Aug 2025 – May 2027

B.S. in Mechanical Engineering *GPA: 4.0/4.0*

Aug 2022 – May 2025

Awards: CIT Leadership and Service Award; CIT Honor Research; SURF Spotlight Researcher

Relevant Coursework: Optimal Control and Reinforcement Learning; Robot Learning; Feedback Control Systems; Soft Robotics Modeling; Robotics & AI for Agriculture; Mobile Robots; Legged Systems.

EXPERIENCE

Physical AI Engineer Intern

Summer 2026

Stealth Robotics Startup, Boston, MA

- Developed **reinforcement-learning** control policies in **NVIDIA Isaac Lab** with **RSL-RL**, replacing a model-based QP/IK posture controller on a multi-DOF mobile platform, designed for **sim-to-real** transfer.

RESEARCH

Self-Stabilizing Penguin-inspired Bipedal Robot for Low-Friction Slopes

May 2025 – Present

Research Assistant, Robomechanics Lab (PI: Prof. Aaron Johnson)

- Modeled and fabricated a 3D model of a 5-DOF bipedal robot in **SolidWorks**, implementing a **crank-link leg extension system** that captures macaroni penguin biological inertial properties for energy-efficient waddling.
- Developed a **CPG-RL** framework to create stable, biomimetic walking gaits; integrated **closed-chain constraints** to optimize locomotion on low-friction, inclined surfaces.
- Developed **digital twins** in **MuJoCo** and **NVIDIA Isaac Lab** to validate contact-rich dynamics and closed-loop control policies in challenging environments.
- Developed an experimental testbed for evaluating locomotion and conducted experiments in a **motion capture laboratory** on viscous and slippery substrates.
- **Target Publication:** Lead author paper submission and presentation for **ICRA 2027** (Seoul).
- **Publication:** **Co-First Author** Workshop presentation at **ICRA 2026** (Vienna).
- **Demo:** Live hardware walking demonstration at the NFL Draft Tech Demo Day, Pittsburgh 2026.

Design and Control for Mugatu: A Minimally-Actuated Steerable Bipedal Robot

Dec 2023 – Present

Research Assistant, Robomechanics Lab (PI: Prof. Aaron Johnson)

- Engineered a **PID attitude controller** for the first steerable single-motor biped, enabling directional control.
- Conducted **2D contact dynamics** analysis in MATLAB to investigate **limit cycle** stability and optimize gait efficiency.
- Validated non-linear dynamics and steering manifolds using **Drake** and **MuJoCo** to bridge the gap between theoretical stability, simulation and physical execution.
- **Publication:** **Allometric Scaling Laws for Bipedal Robots**, submitted to **IROS 2026** (arXiv: 2603.22560). Presented at **Gordon Research Conference on Robotics 2026**, **NCUR 2025** and **CMU MOTM 2025**.

PROJECTS

Hybrid Locomotion for Unitree Go2-W in Agriculture Environments

Jan 2026 – Present

Course Project, Robotics & AI for Agriculture (PI: Prof. George Kantor & Francisco Yandun)

- Implementing a **PPO-based RL policy** in **NVIDIA Isaac Lab** for adaptive wheel-leg switching on the Go2-W.
- Training robust navigation in procedurally generated off-road terrains with multi-crop row constraints.

MPPI-Based Autonomous High-Speed Drone Racing

Jan 2026 – May 2026

Course Project, Mobile Robots (PI: Prof. Wennie Tabib)

- Developed an **MPPI controller** for a 17-state quadrotor model to achieve real-time, aggressive gate navigation.
- Optimized trajectories in PyBullet by sampling control inputs to balance high velocity and collision avoidance.

Sim-to-Real RL: Autonomous Quadrotor Landing

Aug 2025 – Jan 2026

Course Project, Robot Learning (PI: Prof. Guanya Shi)

- Synthesized a **PPO controller** with 4096 parallel agents and designed a curriculum learning scheme with exp reward scaling for precision landing on moving targets (<4 cm error).
- Mitigated the sim-to-real gap via **domain randomization**, **96% success rate** on Crazyflie 2.1+ hardware.

SKILLS

- **Programming:** Python, C++, C, MATLAB, Julia, Assembly, Bash, LaTeX
- **Robotics & Simulation:** Isaac Lab, MuJoCo, Drake, MJPC, Dojo, ROS2, Reinforcement Learning (PPO, DQN, CPG), Optimal Control (LQR, MPC, MPPI)
- **Design, Manufacturing, & Analysis:** SolidWorks, AutoCAD, Blender, OnShape, ANSYS Workbench, FEA, GibbsCam, MeshLab, CNC, Mill, Lathe, Laser Cutter, Water Jet, 3D Printing